Understanding the Cost of Compression: BERT vs. DistilBERT

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Introduction

Motivation: Large pretrained models like BERT achieve high performance but carry prohibitive **computational and memory costs** for resource-constrained settings.

The DistilBERT Solution: A compressed version of BERT trained via **knowledge distillation to be smaller and faster** while retaining core language understanding.

Performance Claims: The original paper reports DistilBERT is **40%** smaller, **60%** faster, and retains **~97%** of BERT's performance.

Research Goal: Quantify the specific "cost of compression" by **identifying** where **accuracy drops** in downstream **classification**.

Hypothesis: DistilBERT's loss is concentrated in tasks requiring **compositional logic** (negation) and **long-range dependencies**.

Methodology

We compare model predictions on individual examples to characterize qualitative failure modes, such as negation and long-context reasoning.

- **Datasets**: IMDb, SST-2, MRPC
- **Models**: BERT-base and DistilBERT. DistilBERT is a distilled student model that preserves the encoder-only Transformer design of BERT while using fewer layers.
- **Tasks**: Downstream sentiment classification on SST-2 and IMDb.
- **Evaluation**: Accuracy, F1, training time, inference time, and parameter count.
- **Analysis**: Prediction-level outputs, confidence scores, and text-length statistics are used to study where DistilBERT loses performance relative to BERT.

Results

DistilBERT is smaller and usually faster than BERT, while keeping similar performance. It is slightly worse on SST-2 and IMDb, but slightly better on MRPC.

Paper claim	Our measurement	Trend
40% smaller	39% smaller (67.0M / 109.5M params)	✓
60% faster (training)	30% (SST-2) / 48% (IMDb) / 40% (MRPC) faster	Partial
Retains ~97% of capability	96.9% (SST-2) / 99.1% (IMDb) / 101.8% (MRPC) of BERT's accuracy	✓+

	SST-2 (872 val)		IMDb (25,000 test)		MRPC (408 val)	
	BERT	DistilBERT	BERT	DistilBERT	BERT	DistilBERT
Accuracy	**0.928**	0.899	**0.924**	0.916	0.821	**0.836**
F1	**0.930**	0.902	**0.925**	0.916	0.876	**0.883**
Accuracy gap (pp)		-2.87		-0.84		+1.47
Training time	391 s	275 s	1013 s	530 s	52 s	31 s
Training speed-up		1.42x		1.91x		1.65x
Parameters	109.5M	67.0 M	109.5 M	67.0 M	109.5 M	67.0 M
Inference latency (ms / sample)	2.24	1.98	3.48	1.87	7.62	8.31
Inference speed-up		1.13x		1.86x		0.92x
Agreement rate		95.3%		95.2%		86.3%

Discussion

Systematic Performance Gaps

- **Task-Dependent Accuracy:** Gap is wider on short sentences (SST-2: **-2.87 pp**) than long documents (IMDb: **-0.84 pp**).
- **The "MRPC Paradox":** DistilBERT outperforms BERT (**+1.47 pp**); reduced capacity prevents over-fitting to lexical overlap.
- **Inconsistent Speed-up:** Training is consistently faster (**1.4x–1.9x**), but less benefit on short inputs (IMDb: **1.86x** vs. SST-2: **1.13x**).
- **Altered Boundaries:** Agreement drops from **~95%** (sentiment) to **86.3%** (MRPC), proving distillation changes complex relational logic.

Where DistilBERT Fails

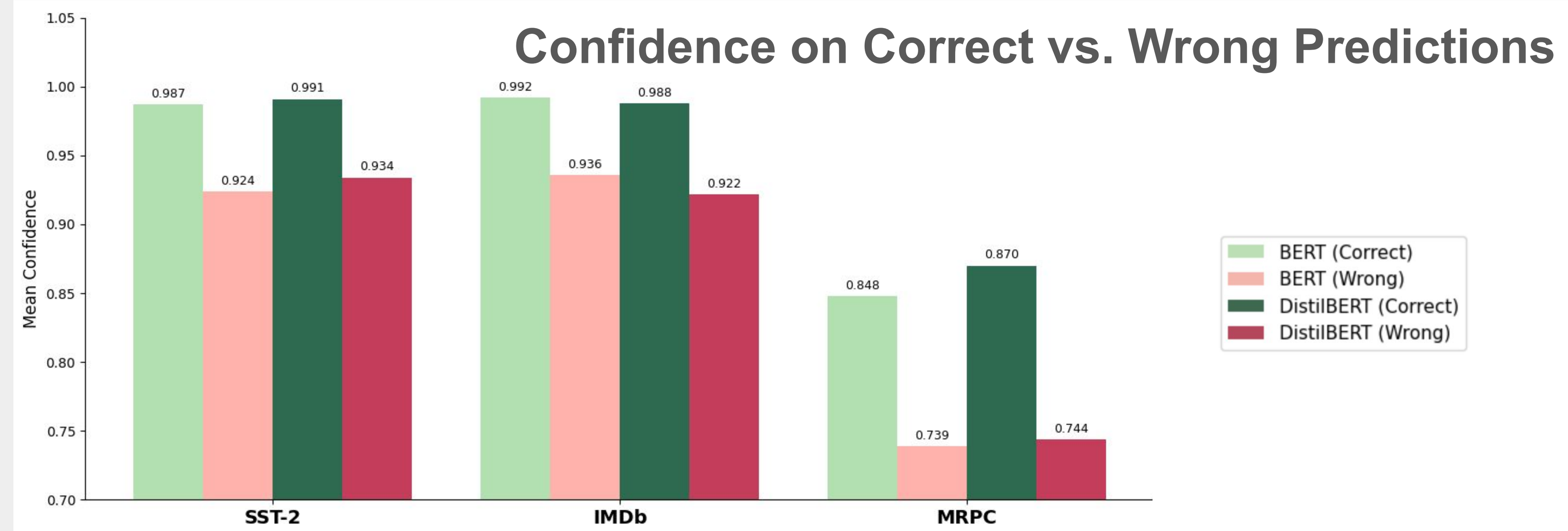
💭 Linguistic Complexity	📜 Long Doc Reasoning	😕 Confidence
when sentiment depends on structure: • Negation ("don't like") • Contrast ("good but bad") • Sarcasm / idioms ➜ relies on surface words instead of structure	long reviews e.g. IMDB • mixed sentiment • delayed conclusions • opinion shifts ➜ overweights early signals ➜ misses final judgment	Overconfident when wrong ➜ poor uncertainty estimation

Key Insights

1. Distillation ≠ performance loss; ➜ changes decision boundary
2. DistilBERT sometimes wins (contrast cases)
3. Outperforms BERT on MRPC ➜ less overfitting
4. Acts as regularization (low-data settings)

"I didn't think it was going to be this good."

BERT: positive! 🙂
DistilBERT: negative! 😈



Conclusion

Validation of Claims: Confirmed DistilBERT retains **97–99% accuracy** while providing **1.4x–1.9x faster training**.

Contextual Loss: Compression costs are not uniform; performance drops significantly on tasks requiring **compositional logic** or **long-range dependencies**.

The MRPC Paradox: Reduced capacity acts as a **regularizer** in low-data settings, allowing the student to outperform the teacher.

Deployment Risks: High accuracy belies **poor calibration**; DistilBERT is "confidently wrong," making it unsuitable for safety-critical tasks.

Future Work

💪 **Robustness Distillation:** Incorporate **contrastive learning or saliency matching** to better capture negation and sarcasm.

🤨 **Uncertainty Calibration:** Apply **Scaling or Smoothing** techniques during distillation.

🐾 **"In-the-Wild" Analysis:** Compare BERT-DistilBERT gaps on **noisy, non-benchmark** data.

👀 **Architecture Comparison:** Investigate whether long-context reasoning failures persist **across other "shallow" models**.

References

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